

Notes on Reinforcement Learning for Quantum Sensing: Phase and Field Estimation Across Quantum Platforms

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Reinforcement learning (RL) has emerged as a powerful paradigm for the optimization of quantum sensors operating in complex, noisy, and time-dependent environments. This article reviews the application of RL to quantum sensing tasks with an emphasis on phase and field estimation. We focus on platforms relevant to contemporary quantum technology, including quantum dots, trapped ions, and neutral atoms. The theoretical framework combines Bayesian inference, Fisher information, and partially observable Markov decision processes (POMDPs), allowing RL agents to adaptively design probe states, control sequences, and measurement strategies. We discuss model-free and model-aware RL approaches, provide pseudocode for representative algorithms, and analyze practical considerations for experimental integration. The results demonstrate how RL can autonomously approach quantum-limited sensitivity and offers a unifying methodology for next-generation quantum metrology.

I. INTRODUCTION

Quantum sensing exploits the coherent dynamics of quantum systems to estimate physical parameters such as phases, frequencies, and fields with extreme precision. In idealized scenarios, the achievable precision is bounded by the quantum Cramér–Rao inequality, which is determined by the quantum Fisher information (QFI) of the probe state. In practice, however, realistic sensors must operate under noise, decoherence, control imperfections, and incomplete information about the parameter of interest.

A canonical sensing task is phase estimation: preparing a quantum state, letting it pick up an unknown phase ϕ , then measuring to infer ϕ . The ultimate precision is limited by the Cramér–Rao bound, $\text{Var}(\hat{\phi}) \geq 1/(NF)$, where F is the Fisher information and N the number of trials. In quantum metrology, one maximizes the quantum Fisher information (QFI) of the probe state to approach the quantum Cramér–Rao limit. For a pure probe state $\psi(\phi)$

$$F_Q(\phi) = 4 \left(\langle \partial_\phi \psi | \partial_\phi \psi \rangle - |\langle \psi | \partial_\phi \psi \rangle|^2 \right),$$

which sets a lower bound $\text{Var}(\hat{\phi}) \geq 1/[NF_Q(\phi)]$. In practice, one performs adaptive measurements and Bayesian updating to approach this bound. For example, given a prior distribution $P(\phi)$ and a measurement outcome m with likelihood $P(m|\phi)$, Bayesian inference yields the posterior $P(\phi|m) \propto P(m|\phi)P(\phi)$. Classical Fisher information for a measurement $p_k(\phi)$ is

$$F_C(\phi) = \sum_k \frac{1}{p_k(\phi)} \left(\frac{\partial p_k(\phi)}{\partial \phi} \right)^2,$$

while the QFI is the maximal F_C over all measurements.

Traditional quantum control methods (GRAPE, CRAB) can optimize fixed sensing tasks, but become expensive or fragile for complex, noisy, time-dependent

scenarios. Machine learning offers an alternative: RL can train an agent to design control protocols or adaptive measurement policies through trial-and-error. Recent works have demonstrated deep RL (policy gradients, Q-learning, cross-entropy) for quantum control and sensing, often exceeding conventional strategies. For example, RL-optimized control pulses in a “quantum-chaotic” sensor were shown to reduce decoherence effects and improve sensitivity by an order of magnitude. Model-aware RL (using gradient information via automatic differentiation) has also been applied, achieving state-of-the-art Bayesian experimental design across platforms. This report expands on these approaches for specific sensor platforms and provides practical guidance for experimenters.

Adaptive sensing strategies based on Bayesian inference have proven effective in approaching quantum limits, but the design of optimal control and measurement policies quickly becomes intractable as the complexity of the sensor increases. Reinforcement learning provides a natural framework for addressing this challenge by enabling an agent to learn adaptive strategies through interaction with the sensing environment.

This work surveys RL-based approaches to quantum sensing, emphasizing phase and field estimation and highlighting concrete implementations in quantum dots, trapped ions, and neutral atom systems.

II. BAYESIAN UPDATES AND POMDP FORMULATION

Adaptive quantum sensing can be formulated as a partially observable Markov decision process (POMDP). The unknown parameter θ is treated as a hidden state, while the agent maintains a belief distribution $b(\theta)$ updated via Bayes’ rule. The agent’s belief state is the Bayesian posterior $b(\theta)$ over θ . Actions a correspond to experimental controls (choice of probe state, evolution time, control Hamiltonians, or measurement basis), and observations

o are measurement outcomes. The transition dynamics are trivial (the parameter does not change between measurements), but each action a yields an outcome o with probability $P(o|\theta, a)$. After observing o , the belief is updated via Bayes' rule:

$$b'(\theta) = \frac{P(o|\theta, a) b(\theta)}{\int d\theta' P(o|\theta', a) b(\theta')}, \quad (1)$$

where a denotes a control action and o a measurement outcome.

The reward function is typically chosen to reflect information gain or precision, such as the reduction in posterior variance or entropy. Representing the belief state explicitly converts the POMDP into a fully observable belief-space MDP, enabling the use of standard RL algorithms.

III. FISHER INFORMATION AND QUANTUM LIMITS

For a measurement distribution $p(o|\theta)$, the classical Fisher information is

$$F_C(\theta) = \sum_o \frac{1}{p(o|\theta)} \left(\frac{\partial p(o|\theta)}{\partial \theta} \right)^2. \quad (2)$$

The quantum Fisher information F_Q provides an upper bound on F_C over all possible measurements. For pure states,

$$F_Q = 4 \left(\langle \partial_\theta \psi | \partial_\theta \psi \rangle - |\langle \psi | \partial_\theta \psi \rangle|^2 \right). \quad (3)$$

RL policies can be trained to maximize rewards related to Fisher information, thereby driving the sensing protocol toward quantum-limited performance.

IV. REINFORCEMENT LEARNING METHODS

A. Model-Free Reinforcement Learning

Model-free RL methods (e.g. Q-learning, Deep Q-Networks, Policy Gradient/A3C, Proximal Policy Optimization) treat the sensor as a black-box. The agent learns via trial outcomes and rewards, without explicit knowledge of the system model. Such methods can handle highly nonlinear, noisy dynamics and accommodate arbitrary reward functions. However, they may require large numbers of episodes to converge, which can be a limitation in experimental settings. For example, earlier RL approaches to phase estimation achieved good results but needed many iterations.

B. Model-Aware Reinforcement Learning

Model-aware RL incorporates known physics into the learning. If a (differentiable) model of the sensor dynamics is available, one can backpropagate gradients through

the simulation to optimize a policy network. Belliaro et al. demonstrated that such gradient-based RL combined with particle-filter Bayesian updates can automatically design near-optimal experimental sequences across platforms. Model-aware methods can directly optimize a reward defined in terms of Fisher information or estimation variance. They also allow leveraging analytical gradients to speed up training.

C. Offline RL and Domain Randomization

Offline RL and Domain Randomization: In many quantum experiments, data are expensive or time-consuming to acquire. Offline RL (learning from a fixed dataset) can exploit pre-collected measurements or simulated data to train policies without active experimentation. Domain randomization extends this by intentionally varying simulation parameters (noise levels, decoherence rates, system inhomogeneities) during training, so that the learned policy is robust to model mismatch. For instance, one might randomize a simulated magnetic field or phase shift to ensure the agent performs well under calibration uncertainties. While specific work on offline RL in quantum sensing is still emerging, these techniques from robotics and control could be adapted to quantum setups.

V. RL IMPLEMENTATION BLUEPRINT

An actual implementation would follow these steps:

1. **Define the environment:** Choose the sensor platform and identify its controllable parameters and observables. Build a simulation of the sensor dynamics (e.g. master equation or Monte Carlo) that includes the unknown parameter. If possible, validate the simulation against experiments.
2. **State representation:** Decide what information to feed to the agent. This may be raw data (fluorescence images, spin readouts) or pre-processed features (means and variances). Include memory of previous steps if needed (e.g. via recurrent networks) to handle non-Markovian effects.
3. **Action space:** Specify the set of actions. For continuous controls, discretization or parameterized actions can be used. Ensure actions respect experimental limits (e.g. maximum pulse amplitude).
4. **Reward function:** Choose a reward that captures precision. A terminal reward $R = -\log(\text{estimation variance})$ is one option. Alternatively, use stepwise rewards like Bayesian information gain. Regularize the reward if necessary to avoid trivial strategies.
5. **RL algorithm:** Select an algorithm based on action type. For example, use DQN for coarse adjustment

of discrete parameters, or SAC/DDPG for smooth continuous control. Set neural network architectures appropriate for the state (e.g. convolutional nets for images).

6. **Training:** Train the agent in simulation. Introduce variability (randomize unknowns) to encourage generalization. Use experience replay or batch updates to improve sample efficiency. Monitor performance on validation scenarios.
7. **Deployment:** Transfer the trained policy to the real device. If training was in-silico, consider fine-tuning on the hardware with a few adjustment episodes. Include safeguards for safety (e.g. bounding pulses) and mechanisms to reset the environment.

VI. QUANTUM SENSING PLATFORMS

A. Quantum Dot Sensors

Semiconductor quantum dot devices can act as tunable charge or spin sensors for electric and magnetic fields. Double quantum dots (DQDs) are widely used for qubit readout and have complex stability diagrams. Deep RL has been applied to automate DQD measurement: Nguyen et al. demonstrated a DQN-based agent that rapidly locates “bias triangles” in a transport map of a GaAs double dot . This amounts to learning an experimental protocol (gate voltages to apply and where to measure) to identify a narrow feature in a large parameter space. Their algorithm found triangles in minutes, far faster than manual tuning . This approach shows that RL can replace expert heuristics in navigating dot operation regimes. Similar RL agents could be trained to adaptively measure phase shifts in a quantum dot interferometer or to optimize gate voltages for maximum charge sensitivity.

In practice, one would model the dot sensor’s output o (current or charge) as a function of the unknown parameter ϕ (e.g. a phase imposed on the electron spin) and the control voltages a . The agent then updates its belief on ϕ after each measurement. Since dot charge sensors often have limited quantum coherence, these tasks can often be well described as classical POMDPs, though spin qubits in dots could require quantum updates. The RL policy might be a neural network taking the current belief and possibly past history as input and outputting the next gate voltage configuration. Domain randomization can account for device variability between cooldowns. In short, RL offers a flexible framework for autonomous tuning and adaptive estimation in quantum dot sensors .

B. Trapped Ion Sensors

Ion traps confine individual atomic ions (e.g. Yb⁺, Ca⁺) using electric and/or magnetic fields. These plat-

forms are used for quantum clocks, electric-field sensing, and magnetometry. The control parameters include laser frequencies/pulses and voltages on segmented electrodes. The challenges include compensating stray fields and designing pulse sequences under decoherence. Recent studies have applied ML and RL in this context. For example, Schmitz et al. used adaptive algorithms to optimize stray electric field compensation for a ¹⁷¹Yb ion. *They compared gradient descent (ADAM) with a machine-learning optimizer (MLOOP) to maximize ion fluorescence. Starting with a 44-dimensional control (44 DC electrodes) to maintain optimal sensor operation.*

In this setup, an RL agent could treat the set of compensation voltages and/or laser beam parameters as the action space. For parameter estimation (e.g. sensing a phase imprinted on the ion’s internal state), the agent would iteratively apply a control (like a rotation angle) and measure fluorescence, updating a phase posterior. Cross-entropy RL and policy-gradient methods have been used to find optimal pulse sequences in trapped-ion sensing. Schuff et al. demonstrated that RL (cross-entropy method) can discover spin-squeezing-like pulse strategies in a collective-spin ion sensor, outperforming simple periodic pulses . Importantly, the agent learned to mitigate decoherence by adjusting pulse timing and strength, yielding up to an order-of-magnitude sensitivity gain . This adaptive RL protocol effectively enhances the quantum Fisher information of the sensing scheme.

In implementation, one must consider experimental constraints: real-time feedback bandwidth, laser noise, and ion heating. Offline RL or simulation-based pretraining can reduce the number of costly live trials. For instance, by simulating the ion’s master-equation dynamics, an agent can learn a policy before deployment, then fine-tune on the actual hardware. Domain randomization (varying decoherence rates or trap frequencies in simulation) can make the policy robust to model mismatch. Integration with ion-trap software (e.g. ARTIQ or custom FPGA control) allows the agent to select pulses or compensation voltages on the fly. Careful reward shaping (e.g. using reductions in state variance or improvements in fluorescence) will guide the RL to physically meaningful sensing strategies .

C. Neutral Atom Sensors

Neutral-atom platforms include cold-atom interferometers and magnetometers. Alkali vapor cells (e.g. Rb, Cs) and ultracold Bose–Einstein condensates (BECs) can serve as precision sensors for fields and acceleration. A recent cutting-edge example is using ultracold atoms in an optical lattice as an inertial sensor. Chih and Holland applied reinforcement learning to design a novel atom-lattice gyroscope . Instead of the standard Mach–Zehnder sequence, the RL agent learned an end-to-end “shaking” protocol for a 2D lattice. Remarkably, for the same total interrogation time, the learned protocol achieved a 20-fold improvement in rotation sensitivity over conventional in-

Algorithm 1 RL-Based Adaptive Phase Estimation

```

1: Initialize belief  $b(\phi)$ , policy  $\pi$ 
2: for each episode do
3:   for  $t = 1$  to  $T$  do
4:     Select action  $a_t = \pi(b(\phi))$ 
5:     Perform measurement, observe  $o_t$ 
6:     Update belief  $b(\phi)$  via Bayes' rule
7:     Compute reward  $r_t$ 
8:     Update policy  $\pi$ 
9:   end for
10: end for

```

terferometry . This demonstrates RL's ability to discover non-intuitive quantum sensing schemes by maximizing long-term rewards.

Neutral atoms also include solid-state ensembles like NV-center magnetometers, which can be treated similarly. An agent could adaptively choose microwave pulses and measurement bases to track a time-varying magnetic field, updating a field estimate after each measurement. Bayesian RL approaches (policy gradient with belief updates) could greatly speed up such magnetometry.

On Earth, atomic ensembles in magneto-optical traps or lattice arrays act as vector magnetometers and clocks. Reinforcement learning can optimize these too: for example, an RL agent might learn how long to interrogate a Ramsey sequence based on the current estimate of an oscillating field, effectively learning adaptive control that approaches the quantum projection noise limit. Experiments have begun to implement RL for multi-atom systems; one may adapt existing RL toolkits (e.g. `qsensoropt`) to these platforms. Key practical considerations include dealing with large particle numbers and decoherence. Fortunately, the notion of QFI extends to many-body states, and RL can be used to approach the best possible scaling (shot-noise or beyond). Control actions may include global phase shifts (via magnetic fields) and collective readout choices. Reinforcement learning can thus serve as a unifying framework for many neutral-atom sensing modalities .

VII. PSEUDOCODE EXAMPLES

To illustrate how one might implement an RL-based adaptive estimator, we give pseudocode for two representative tasks.

```

1: Algorithm: RL-based Adaptive Phase Estimation
2: Initialize prior distribution  $P(\phi)$  uniform over  $[0, 2\pi)$ , policy network (or Q-table)  $\pi$ , and replay buffer (if applicable).
3: for episode = 1 to  $N_{\text{ep}}$  do
4:   Reset belief  $b(\phi) \leftarrow P(\phi)$ .
5:   for step = 1 to  $T$  do
6:     Action: choose control parameter  $a_t = \pi(b(\phi))$  (e.g. phase shift or pulse duration).

```

```

7:     Probe: apply  $a_t$  to sensor, obtain measurement outcome  $m_t \in \{0, 1\}$ .
8:     Belief update: use Bayes:  $b(\phi) \propto P(m_t|\phi, a_t) b(\phi)$ , then normalize.
9:     Reward: compute reward  $r_t$ , e.g.  $-\text{Var}_b(\phi)$  or information gain  $D_{\text{KL}}(b \| b_{\text{prior}})$ .
10:    Store (state =  $b(\phi)$ , action =  $a_t$ ,  $r = r_t$ , next state =  $b'(\phi)$ ).
11:    Update policy  $\pi$  (Q-learning or gradient step) using the transition and reward.
12:  end for
13: end for

```

The above loop exemplifies adaptive Bayesian estimation: after each measurement, the agent updates the posterior $b(\phi)$ and uses it as input for the next action. This fully Bayesian RL approach is data-efficient, reusing all prior information . By shaping the reward with estimation variance, the agent learns to drive the posterior toward a narrow distribution, effectively maximizing Fisher information.

```

1: Algorithm: RL for Magnetic Field Estimation with NV Centers
2: Initialize initial state estimate  $b(B)$  (over magnetic field  $B$ ), policy  $\pi$ , and environment model (optional).
3: for episode = 1 to  $N_{\text{ep}}$  do
4:   Reset belief  $b(B)$  to prior (e.g. broad Gaussian).
5:   for step = 1 to  $T$  do
6:     Action: select control sequence  $a_t$  (e.g. microwave pulse amplitude and duration, or Ramsey time).
7:     Apply: implement  $a_t$  on the NV center, record photoluminescence signal  $m_t$ .
8:     Likelihood: compute  $P(m_t|B, a_t)$  from the sensor model.
9:     Belief update:  $b(B) \propto P(m_t|B, a_t) b(B)$ .
10:    Reward:  $r_t = -\text{Var}_b(B)$  (or other metric).
11:    Store transition and update policy  $\pi$ .
12:  end for
13: end for

```

In both examples, the state of the MDP is the current belief distribution (or a finite representation of it) and the observation is the latest measurement outcome. These pseudocode sketches can be implemented with deep neural networks (taking a discretized belief or sufficient statistics as input) or simpler methods if the parameter space is small. They illustrate how RL integrates control choice, Bayesian inference, and reward-driven learning into a closed loop.

VIII. MODEL-FREE VS MODEL-AWARE, OFFLINE RL, DOMAIN RANDOMIZATION

We have noted that model-free RL (e.g. Q-learning) makes minimal assumptions but can be data-hungry. In contrast, model-aware RL (like policy gradient with differentiable simulation) exploits knowledge of the sensor's quantum dynamics. For instance, Bellardo et al. designed

a library (qsensoropt) that uses automatic differentiation through the Lindblad evolution and particle filter to train policies. Their model-aware agent directly computes gradients of precision metrics (Fisher information or posterior covariance) with respect to control parameters, yielding highly efficient training.

Offline RL can leverage pre-recorded experimental data. For example, one might perform a set of calibration measurements offline and train an RL agent on this fixed dataset. Care must be taken to avoid overfitting; conservative algorithms (like Batch Constrained Q-learning) could be adapted. Domain randomization has been crucial in transferring policies from simulation to real robots; similarly, in quantum sensing one can randomize Hamiltonian parameters or noise levels during training. This way, an RL policy learns to work under uncertainty (e.g. unknown decoherence rate). While specific demonstrations in quantum sensing are limited, these methods are straightforward to implement: one simply varies the simulated environment dynamics at the start of each training episode. Doing so can improve robustness when the actual device deviates from the ideal model.

Finally, hybrid methods like Fiderer et al.’s show that evolutionary strategies and RL can jointly optimize adaptive heuristics. In some cases, one could pretrain a neural network to propose near-optimal actions (via supervised learning on simulated “expert” data) and then fine-tune it with RL for final performance. This might mitigate the problem of sparse rewards in experimental data.

IX. IMPLEMENTATION CONSIDERATIONS

Key practical challenges include data efficiency, reward design, hardware constraints, and robustness to noise and drift. Hybrid strategies combining simulation pretraining and experimental fine-tuning are likely to be essential for scalable deployment.

Integrating RL with real quantum sensors requires attention to practical constraints:

- **Data Efficiency:** Experiments are slow. Model-aware training in simulation is recommended, with periodic calibration on real hardware. Agents can use Gaussian process models or Bayesian surrogate models to reduce sample count.
- **Reward Design:** The reward function must reflect sensing goals. Options include negative posterior variance, log-likelihood, or information gain. Explicitly

using quantum-limits (e.g. QFI) in the reward can accelerate learning.

- **Noise and Imperfections:** Real measurements have technical noise, drift, and detection errors. Reward functions should penalize noisy actions. Domain randomization during training can help the agent tolerate such imperfections. One can also include a “no-update” action to handle erasure or no-fluorescence outcomes.
- **Hardware Constraints:** Action space should respect hardware limits (e.g. discrete pulse durations, limited voltage range). Use realistic simulations of the control hardware. An RL policy network or Q-table can run on real-time controllers (FPGA, GPU) to decide the next move between experimental shots.
- **Exploration vs Exploitation:** In an experimental loop, aggressive exploration can degrade performance. A safe strategy is to start with a known good protocol and let RL refine it (e.g. epsilon-greedy around a baseline). Off-policy learning from off-line data can also reduce on-the-fly exploration.
- **Parallel Experiments:** If multiple sensors or multiple runs are possible, parallel data collection can speed up training. Care must be taken if agents share a common policy to avoid correlated failures.

As a concrete example, consider calibrating an atomic magnetometer: an RL agent could sample at different interrogation times to build a field estimate. The pseudocode above shows how each measurement outcome would refine the field belief. One practical recipe is: (1) train the RL policy in a high-fidelity simulator that includes estimated decoherence; (2) load the policy into the experiment and run in “closed loop” with Bayesian updates; (3) continue occasional training online using real outcomes. This kind of mixed-model RL leverages simulations for learning efficiency but corrects for reality with real data.

X. CONCLUSION

Reinforcement learning provides a versatile and powerful framework for quantum sensing. By unifying Bayesian inference, Fisher information, and adaptive control, RL enables autonomous discovery of high-performance sensing protocols across a wide range of quantum platforms. As quantum hardware matures, RL-driven sensing is poised to become a central tool in quantum metrology.